

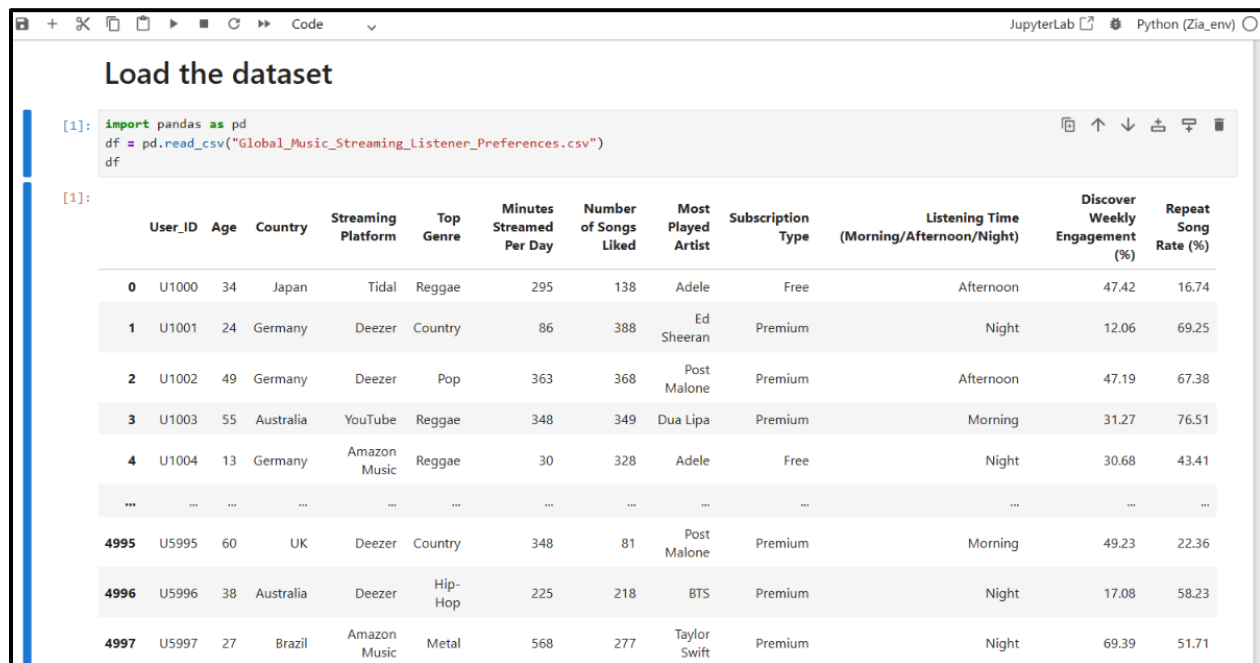
2. Exploratory Data Analysis using Python

Step 1 : Import Libraries and load dataset

- Import libraries like Pandas and Numpy
- Import Visualisation Libraries like Seaborn , Matplotlib

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

- Load dataset using Pandas



The screenshot shows a JupyterLab window with a code editor and a data frame preview. The code cell contains the following Python code:

```
[1]: import pandas as pd
df = pd.read_csv("Global_Music_Streaming_Listener_Preferences.csv")
df
```

The data frame preview shows the following columns: User_ID, Age, Country, Streaming Platform, Top Genre, Minutes Streamed Per Day, Number of Songs Liked, Most Played Artist, Subscription Type, Listening Time (Morning/Afternoon/Night), Discover Weekly Engagement (%), and Repeat Song Rate (%). The preview displays rows 0 to 4, followed by an ellipsis, and then rows 4995 to 4997.

	User_ID	Age	Country	Streaming Platform	Top Genre	Minutes Streamed Per Day	Number of Songs Liked	Most Played Artist	Subscription Type	Listening Time (Morning/Afternoon/Night)	Discover Weekly Engagement (%)	Repeat Song Rate (%)
0	U1000	34	Japan	Tidal	Reggae	295	138	Adele	Free	Afternoon	47.42	16.74
1	U1001	24	Germany	Deezer	Country	86	388	Ed Sheeran	Premium	Night	12.06	69.25
2	U1002	49	Germany	Deezer	Pop	363	368	Post Malone	Premium	Afternoon	47.19	67.38
3	U1003	55	Australia	YouTube	Reggae	348	349	Dua Lipa	Premium	Morning	31.27	76.51
4	U1004	13	Germany	Amazon Music	Reggae	30	328	Adele	Free	Night	30.68	43.41
...	...	...	...	...	...	...	...	...	...	...	...	...
4995	U5995	60	UK	Deezer	Country	348	81	Post Malone	Premium	Morning	49.23	22.36
4996	U5996	38	Australia	Deezer	Hip-Hop	225	218	BTS	Premium	Night	17.08	58.23
4997	U5997	27	Brazil	Amazon Music	Metal	568	277	Taylor Swift	Premium	Night	69.39	51.71

Step 2 : Understanding the Structure and Distribution of Data

- Check the structure and content of the dataset quickly, use `df.head()` .

```
[17]: # check for dataset preview or first five records
df.head()
```

```
[17]:
```

	User_ID	Age	Country	Streaming Platform	Top Genre	Minutes Streamed Per Day	Number of Songs Liked	Most Played Artist	Subscription Type	Listening Time (Morning/Afternoon/Night)	Discover Weekly Engagement (%)	Repeat Song Rate (%)
0	U1000	34	Japan	Tidal	Reggae	295	138	Adele	Free	Afternoon	47.42	16.74
1	U1001	24	Germany	Deezer	Country	86	388	Ed Sheeran	Premium	Night	12.06	69.25
2	U1002	49	Germany	Deezer	Pop	363	368	Post Malone	Premium	Afternoon	47.19	67.38
3	U1003	55	Australia	YouTube	Reggae	348	349	Dua Lipa	Premium	Morning	31.27	76.51
4	U1004	13	Germany	Amazon Music	Reggae	30	328	Adele	Free	Night	30.68	43.41

- Check the concise summary of the dataframe.

```
# checking for summary of dataset
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 12 columns):
#   Column                                                                 Non-Null Count  Dtype
---  -
0   User_ID                                                                5000 non-null   str
1   Age                                                                    5000 non-null   int64
2   Country                                                                5000 non-null   str
3   Streaming Platform                                                     5000 non-null   str
4   Top Genre                                                             5000 non-null   str
5   Minutes Streamed Per Day                                              5000 non-null   int64
6   Number of Songs Liked                                                 5000 non-null   int64
7   Most Played Artist                                                    5000 non-null   str
8   Subscription Type                                                     5000 non-null   str
9   Listening Time (Morning/Afternoon/Night)                             5000 non-null   str
10  Discover Weekly Engagement (%)                                         5000 non-null   float64
11  Repeat Song Rate (%)                                                  5000 non-null   float64
dtypes: float64(2), int64(3), str(7)
memory usage: 689.4 KB
```

It shows that there are no null values in each column and also tells the datatype of each column. We can also check the datatypes by using **df.dtypes**

```
[19]: #checking datatype for each column
df.dtypes
```

```
[19]: User_ID          str
      Age            int64
      Country        str
      Streaming Platform  str
      Top Genre       str
      Minutes Streamed Per Day  int64
      Number of Songs Liked    int64
      Most Played Artist      str
      Subscription Type       str
      Listening Time (Morning/Afternoon/Night)  str
      Discover Weekly Engagement (%)  float64
      Repeat Song Rate (%)  float64
      dtype: object
```

- Check the descriptive statistics of the dataframe.

```
[20]: # checking descriptive statistics
df.describe()
```

```
[20]:
```

	Age	Minutes Streamed Per Day	Number of Songs Liked	Discover Weekly Engagement (%)	Repeat Song Rate (%)
count	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000
mean	36.664000	309.237200	253.51720	50.296464	42.391060
std	13.763844	172.029822	146.36928	23.171936	21.444121
min	13.000000	10.000000	1.000000	10.020000	5.000000
25%	25.000000	161.000000	126.000000	30.147500	24.195000
50%	37.000000	316.000000	254.000000	50.420000	41.960000
75%	49.000000	457.250000	382.000000	70.337500	60.740000
max	60.000000	600.000000	500.000000	89.990000	79.990000

It provides key descriptive statistics for numerical columns (by default), helping you understand the distribution and spread of your data. It shows that the maximum minutes streamed per day is 600 minutes whereas the minimum is 10 minutes , also we can see the average age of users is about 37 .

Step 2 : Perform Data cleaning :

- Check for missing values and handle them appropriately.

```
[21]: #check for missing values
df.isna().sum()

[21]: User_ID          0
      Age             0
      Country         0
      Streaming Platform 0
      Top Genre        0
      Minutes Streamed Per Day 0
      Number of Songs Liked 0
      Most Played Artist 0
      Subscription Type 0
      Listening Time (Morning/Afternoon/Night) 0
      Discover Weekly Engagement (%) 0
      Repeat Song Rate (%) 0
      dtype: int64
```

It shows that there are no null values in any of the columns.

- Remove the duplicate entries.

```
[22]: #checking for duplicates
df.duplicated().sum()

[22]: np.int64(0)
```

It gives the total duplicate entries and as we can see there are no duplicates in the dataset.

- Check the Categorical and Numerical values.

```
[23]: #checking for categorical features
categorical_features = [feature for feature in df.columns if df[feature].dtype == 'string']
categorical_features

[23]: ['User_ID',
      'Country',
      'Streaming Platform',
      'Top Genre',
      'Most Played Artist',
      'Subscription Type',
      'Listening Time (Morning/Afternoon/Night)']
```

Categorical columns in the dataset include **User_id** , **Country**, **Streaming Platform**, **Top Genre**, etc.

```
[24]: # checkin for numerical featutres
numerical_features = [feature for feature in df.columns if df[feature].dtype != "string"]
numerical_features

[24]: ['Age',
      'Minutes Streamed Per Day',
      'Number of Songs Liked',
      'Discover Weekly Engagement (%)',
      'Repeat Song Rate (%)']
```

Numerical Columns in the dataset include **Age, Minutes Streamed Per Day, Number of Songs Liked, etc.**

- Check for unique values in each categorical column

```
[25]: # checking for unique values in each categorical column
for column in categorical_features :
    unique_values = df[column].unique()
    print(f"{column} : {unique_values} ")

User_ID : <ArrowStringArray>
['U1000', 'U1001', 'U1002', 'U1003', 'U1004', 'U1005', 'U1006', 'U1007',
 'U1008', 'U1009',
 ...
 'U5990', 'U5991', 'U5992', 'U5993', 'U5994', 'U5995', 'U5996', 'U5997',
 'U5998', 'U5999']
Length: 5000, dtype: str
Country : <ArrowStringArray>
[ 'Japan',      'Germany',      'Australia', 'South Korea',      'UK',
  'Brazil',     'Canada',       'India',      'USA',             'France']
Length: 10, dtype: str
Streaming Platform : <ArrowStringArray>
['Tidal', 'Deezer', 'YouTube', 'Amazon Music', 'Spotify', 'Apple Music']
Length: 6, dtype: str
Top Genre : <ArrowStringArray>
[ 'Reggae',  'Country',      'Pop',      'EDM', 'Classical',      'Metal',
  'Rock',    'Hip-Hop',      'R&B',      'Jazz']
Length: 10, dtype: str
Most Played Artist : <ArrowStringArray>
[      'Adele',      'Ed Sheeran',      'Post Malone',      'Dua Lipa',
  'The Weeknd', 'Billie Eilish',      'BTS',      'Taylor Swift',
  'Drake',      'Bad Bunny']
Length: 10, dtype: str
Subscription Type : <ArrowStringArray>
['Free', 'Premium']
Length: 2, dtype: str
Listening Time (Morning/Afternoon/Night) : <ArrowStringArray>
['Afternoon', 'Night', 'Morning']
Length: 3, dtype: str
```

The following code shows the unique values in each categorical column. It helps us to see the unique values of each column.

- Removing unnecessary columns (eg. User_ID)

```
[26]: # Removing unnecessary columns (eg. User_ID)
df = df.drop(columns = ["User_ID"])
df
```

```
[26]:
```

	Age	Country	Streaming Platform	Top Genre	Minutes Streamed Per Day	Number of Songs Liked	Most Played Artist	Subscription Type	Listening Time (Morning/Afternoon/Night)	Discover Weekly Engagement (%)	Repeat Song Rate (%)
0	34	Japan	Tidal	Reggae	295	138	Adele	Free	Afternoon	47.42	16.74
1	24	Germany	Deezer	Country	86	388	Ed Sheeran	Premium	Night	12.06	69.25
2	49	Germany	Deezer	Pop	363	368	Post Malone	Premium	Afternoon	47.19	67.38
3	55	Australia	YouTube	Reggae	348	349	Dua Lipa	Premium	Morning	31.27	76.51
4	13	Germany	Amazon Music	Reggae	30	328	Adele	Free	Night	30.68	43.41
...	...	...	...	...	...	...	...	...	...	...	...
4995	60	UK	Deezer	Country	348	81	Post Malone	Premium	Morning	49.23	22.36
4996	38	Australia	Deezer	Hip-Hop	225	218	BTS	Premium	Night	17.08	58.23
4997	27	Brazil	Amazon Music	Metal	568	277	Taylor Swift	Premium	Night	69.39	51.71
4998	58	Canada	Amazon Music	EDM	523	492	Bad Bunny	Premium	Night	43.44	24.53
4999	30	South Korea	YouTube	Rock	451	145	Adele	Premium	Night	15.45	79.46

5000 rows x 11 columns

Since **User_ID** is the unique identifier of each listener so we can drop this column as it will not be too useful if we are doing EDA on the complete dataset but it is useful if we want to analyze each user separately.

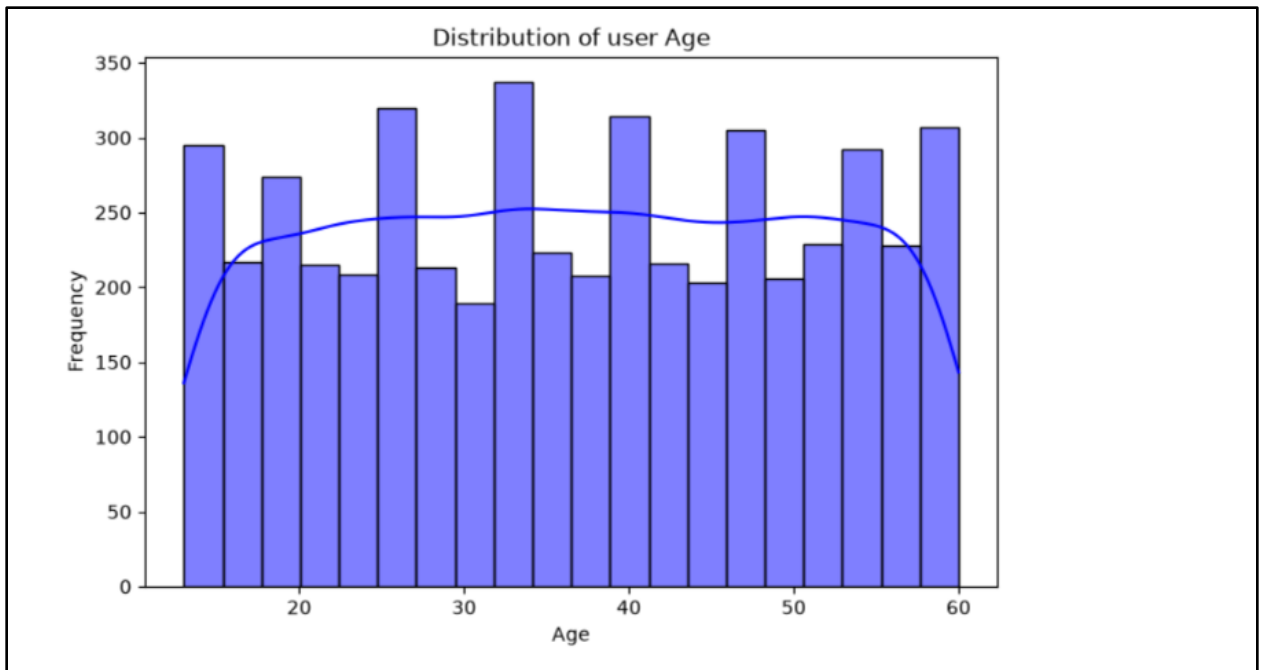
Step 3 : Univariate Analysis

- Distribution of user ages

```
[27]: # Univariate Analysis (Distribution of user ages)

import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

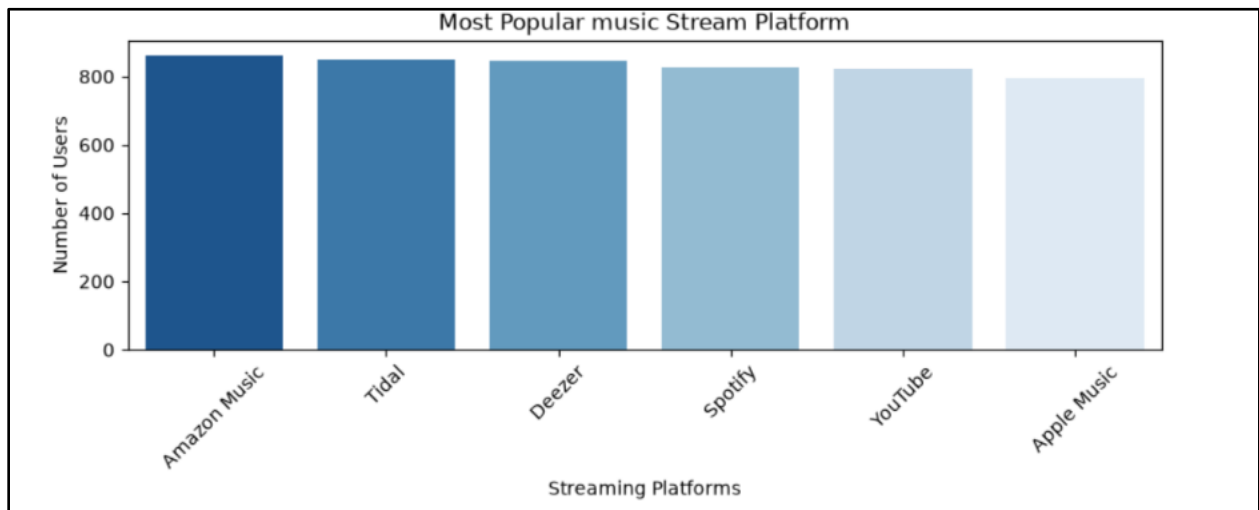
```
[36]: plt.figure(figsize = (8,5))
sns.histplot(df["Age"], bins = 20, kde=True, color="blue")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.title("Distribution of user Age")
plt.show()
```



(Figure 1)

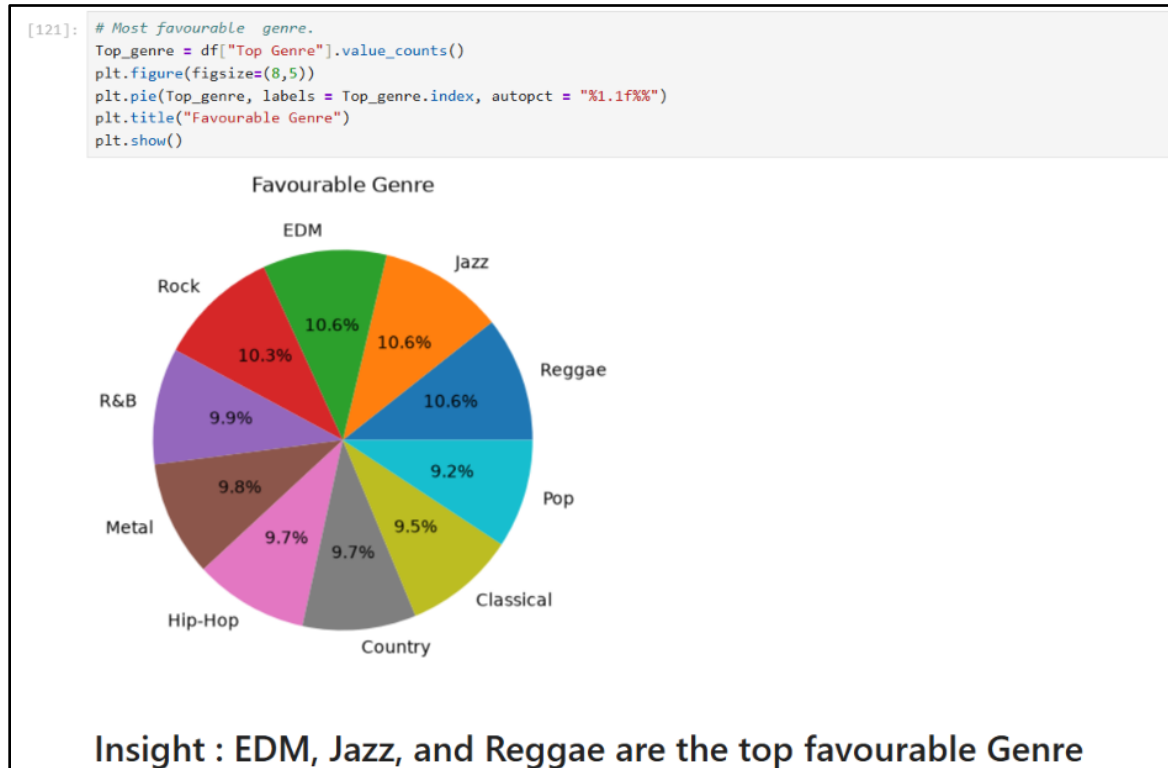
- Popular streaming platform

```
[56]: # Popular streaming platform
plt.figure(figsize=(10,3))
sns.countplot(x= df["Streaming Platform"], order = df["Streaming Platform"].value_counts().index, palette="Blues_r")
plt.xlabel("Streaming Platforms")
plt.ylabel("Number of Users")
plt.title("Most Popular music Stream Platform")
plt.xticks(rotation= 45)
plt.show()
```



(Figure 2)

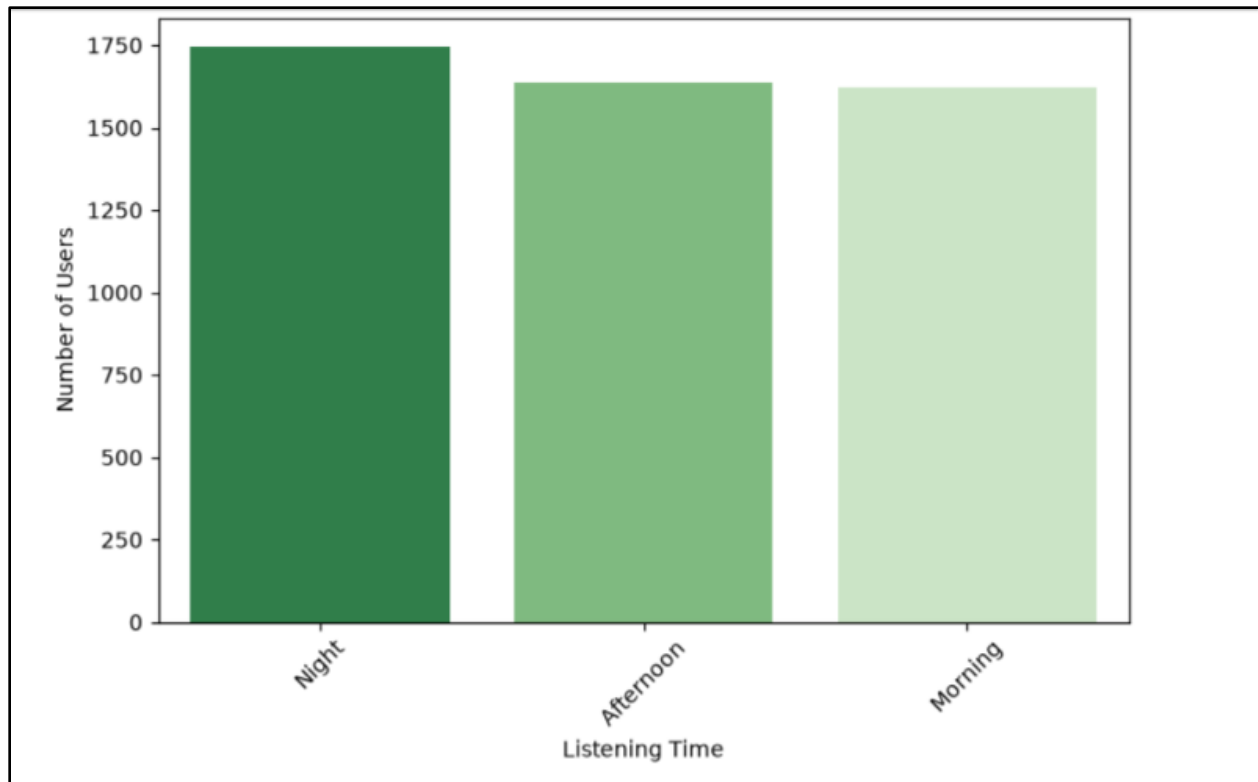
- Most favourable genre.



(Figure 3)

- Favourable time of users to listen songs



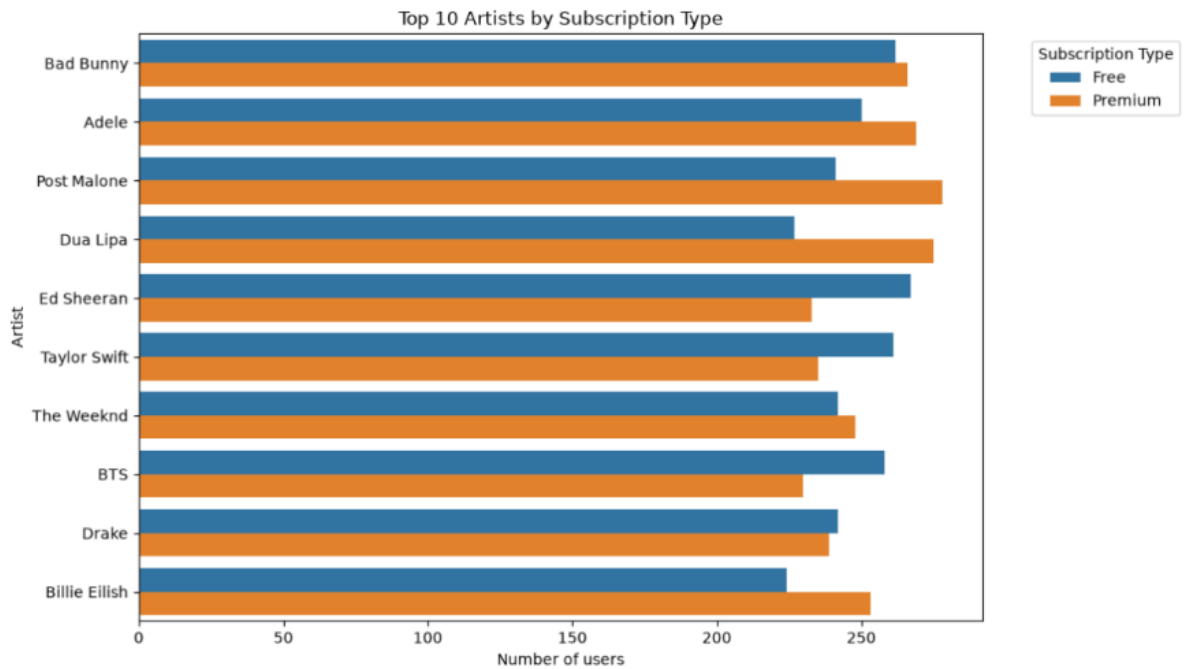


(Figure 4)

Step 4 : Bivariate Analysis and Multivariate Analysis

- **Combo Chart** : Users preferred which subscription type of each artist.

```
[122]: # Bivariate Analysis and Multivariate Analysis (Users preferred which subscription type of each artist.)
plt.figure(figsize = (10,7))
sns.countplot(data = df, y= df["Most Played Artist"], hue="Subscription Type", order=df["Most Played Artist"].value_counts().index[:10])
plt.title("Top 10 Artists by Subscription Type")
plt.xlabel("Number of users")
plt.ylabel("Artist")
plt.legend(title = "Subscription Type", bbox_to_anchor=(1.05,1), loc="upper left")
plt.show()
```



(Figure 5)

- **Heatmap** : Distribution of Music Genres by Age Group

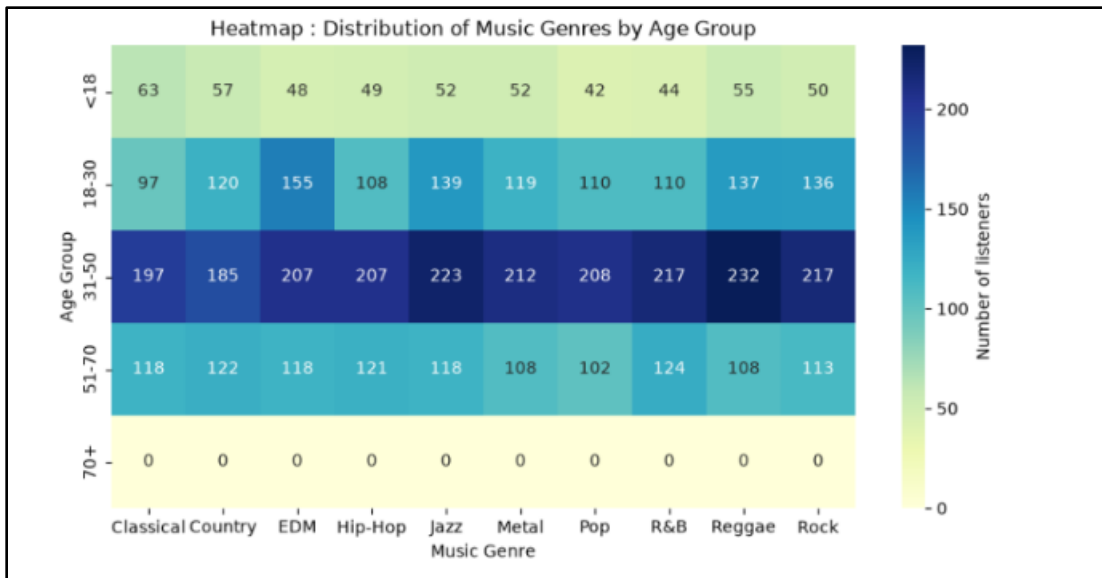
Heatmap : Distribution of Music Genres by Age Group

```
[73]: # Distribution of Music Genres by Age Group

# Creating age groups
bins = [0,18,30,50,70,100]
labels = ["<18", "18-30", "31-50", "51-70", "70+" ]
df["Age Group"] = pd.cut(df["Age"], bins=bins, labels=labels, right=False)

# Grouping data by "Age Group" and "Top Genre"
genre_age_count = df.groupby(["Age Group", "Top Genre"], observed=False).size().unstack().fillna(0)

# creating the heat map
plt.figure(figsize=(10,5))
sns.heatmap(genre_age_count, annot=True, fmt="d", cmap="YlGnBu", cbar_kws= {"label": "Number of listeners"})
plt.title("Heatmap : Distribution of Music Genres by Age Group")
plt.xlabel("Music Genre")
plt.ylabel("Age Group")
plt.show()
```



(Figure 6)

- Relationship between Subscription type and Minutes streamed per day.

```
[78]: # Relationship between Subscription type and Minutes streamed per day.
Subscription_type = df.groupby("Subscription Type")["Minutes Streamed Per Day"].count()
Subscription_type
# Insight : This shows users with premium subscription is more

[78]: Subscription Type
Free      2474
Premium   2526
Name: Minutes Streamed Per Day, dtype: int64
```

(Figure 7)

- Bar Chart** : Relationship between Most Played Artist and Number of Songs Liked.

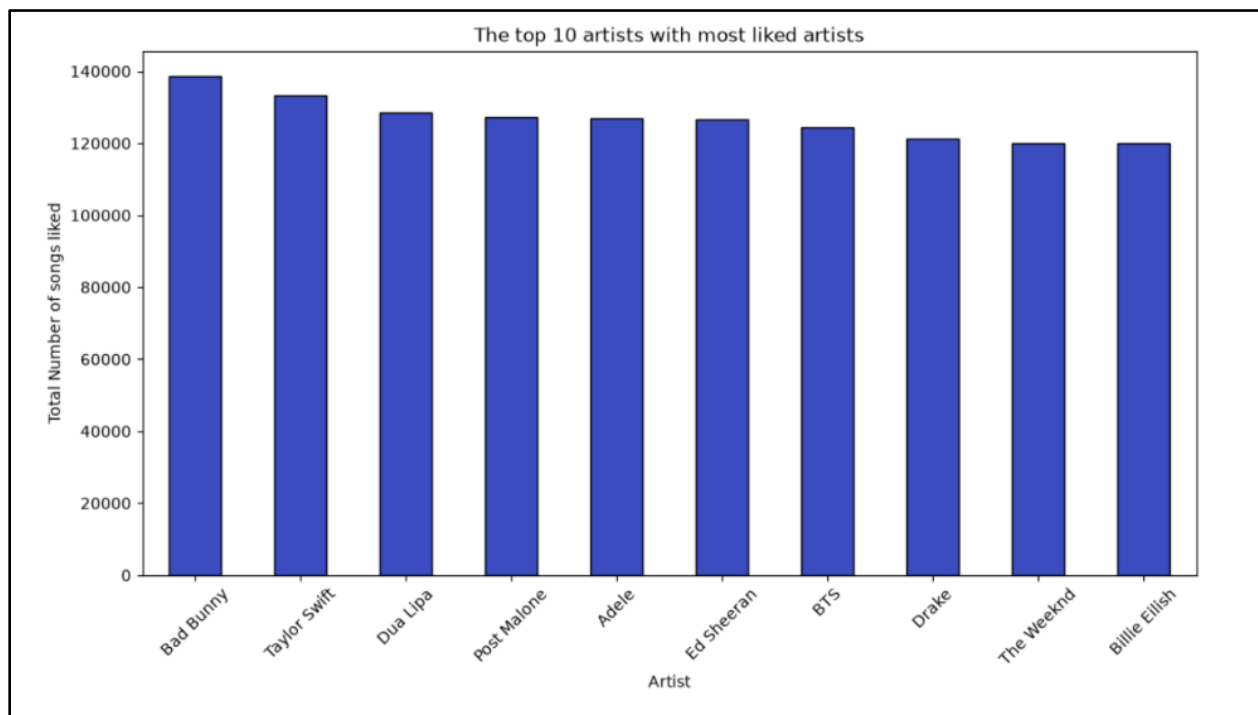
```
[82]: # Relationship between Most Played Artist and Number of Songs Liked.
most_liked_artists = df.groupby("Most Played Artist") ["Number of Songs Liked"].sum().sort_values(ascending=False)

•[81]: # Displaying the top artists with the most liked songs
print(most_liked_artists.head(10))

Most Played Artist
Bad Bunny      138649
Taylor Swift   133319
Dua Lipa       128482
Post Malone    127348
Adele          127072
Ed Sheeran     126640
BTS            124567
Drake          121390
The Weeknd     120134
Billie Eilish  119985
Name: Number of Songs Liked, dtype: int64

[86]: # plotting the top 10 most liked artists
plt.figure(figsize = (12,6))
most_liked_artists.head(10).plot(kind="bar", colormap="coolwarm", edgecolor="black")

# Labels and Titles
plt.xlabel("Artist")
plt.ylabel("Total Number of songs liked")
plt.title("The top 10 artists with most liked artists")
plt.xticks(rotation=45)
plt.show()
```



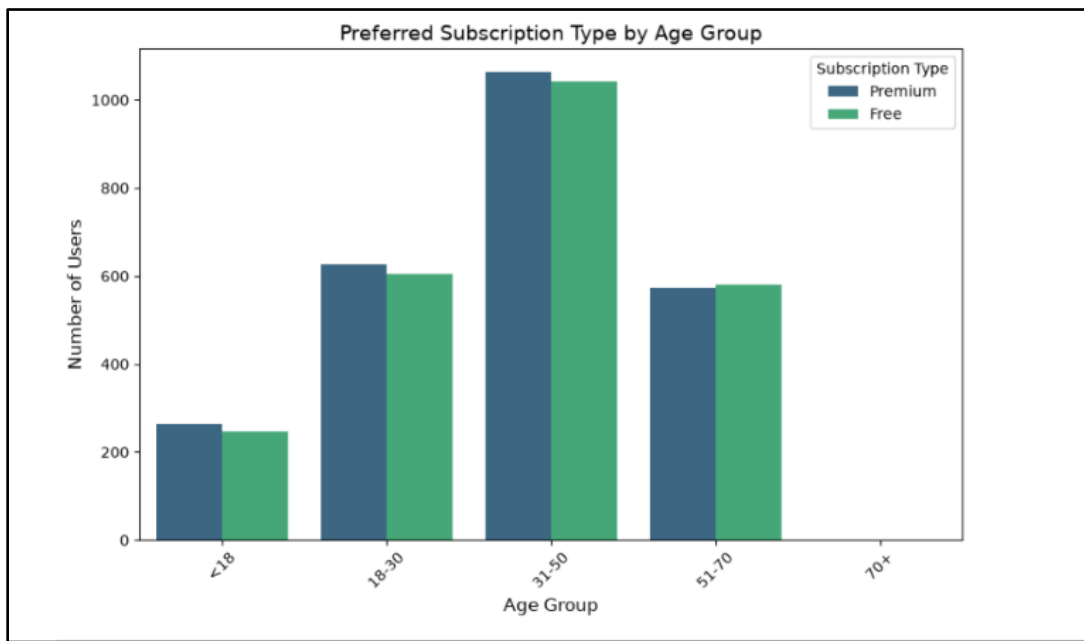
(Figure 8)

- Preferred Subscription Type with Age Group.

Question: What type of subscription do each group prefer ?

```
[92]: # Preferred Subscription Type with Age Group.
# Question: What type of subscription do each group prefer ?
subscription_preference = df.groupby(["Age Group", "Subscription Type"]).size().reset_index(name= "Count")
# sorting values for better visibility
subscription_preference = subscription_preference.sort_values(["Age Group", "Count"], ascending=[True,False])
# set plot style
plt.figure(figsize=(10,6))
sns.barplot(x= "Age Group", y = "Count", hue="Subscription Type",data=subscription_preference,palette="viridis")

#Labels and title
plt.xlabel("Age Group", fontsize=12)
plt.ylabel("Number of Users", fontsize=12)
plt.title("Preferred Subscription Type by Age Group", fontsize=14)
plt.legend(title= "Subscription Type")
plt.xticks(rotation=45)
plt.show()
```



(Figure 9)

Observations from the Chart

1. Distribution of User Ages (Histogram)

- **Result:** The histogram in **Figure 1** shows the age distribution of users. Mild Peaks are in Late 20s & Early 30s. There are slightly higher frequencies around the ages of 28 to 34, suggesting that young adults may be the most engaged demographic. The presence of users ranging from early teens to those in their 50s/60s shows that the platform appeals to a diverse age group.

- **Strategic implications:** It helps streaming services **target content** (e.g., curated playlists or promotions) more effectively by age segment, and also refine **recommendation algorithms** to match user preferences better.

2. Most Popular Streaming Platforms (Bar Chart)

- **Result:** The bar chart in **Figure 2** ranks streaming platforms by the number of users. Amazon Music is the most preferred platform among users in this dataset, closely followed by Tidal and Deezer. The least used platform is **Apple Music**.
- **Strategic implications:** It helps music streaming services understand **user preferences** and can guide **marketing strategies** or **feature prioritization** on specific platforms.
- **Stakeholders:** Competitor analysis teams, platform owners, and marketing strategists.

3. Top Music Genres Preferred (Pie Chart)

- **Result:** From **Figure 3**, we observe EDM, Jazz, and Reggae are the most favored genres, each making up 10.6% of user preferences. Genres like Rock, R&B, and Metal are also relatively popular, while genres such as Pop, Country, and Classical have slightly lower preference rates.
- **Strategic implications:** It helps streaming platforms and music labels identify popular genres and enables platforms to curate personalized playlists based on trending genre popularity.
- **Stakeholders:** Music producers, playlist curators, and record labels.

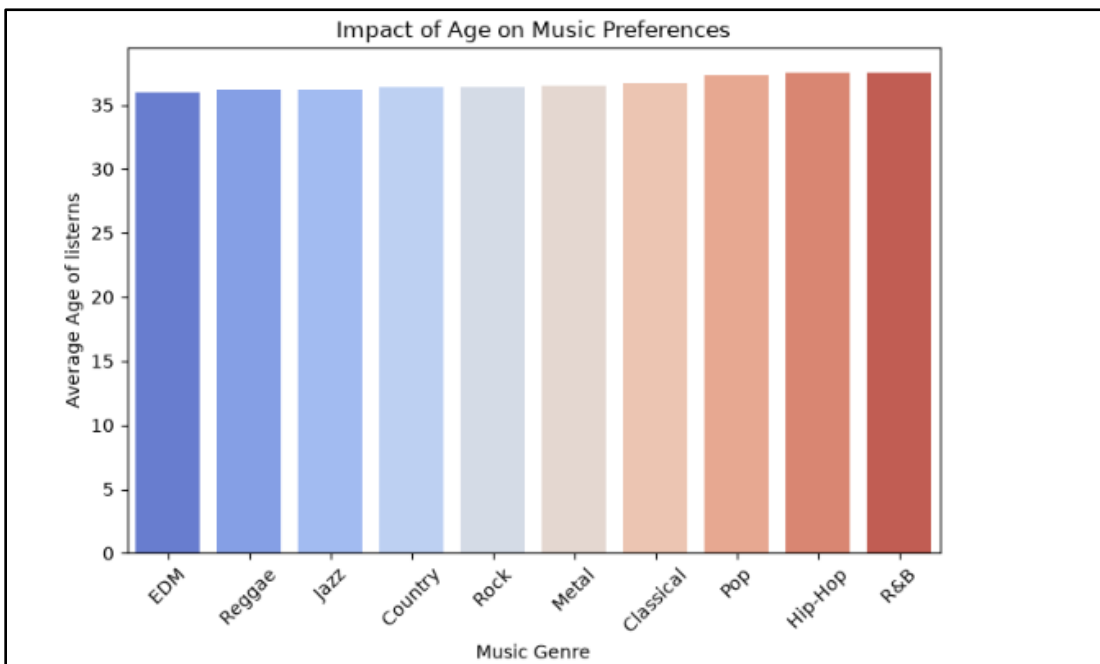
4. Listening Time Preferences (Bar Chart)

- **Result:** From **Figure 4** we observe Night is the most preferred time for listening to music among users. Afternoon and Morning have slightly fewer but nearly equal numbers of listeners.
- **Strategic implications:** This suggests that music streaming platforms could focus on curating and promoting personalized playlists or content more aggressively at night, when engagement is highest.
- **Stakeholders:** Streaming platforms, advertisers, and content creators.

5. Impact of Age on Music Preferences (Bar Chart)

- **Result:** EDM, Reggae, and Jazz are most popular among younger age groups (with the lowest average listener age). Genres like R&B, Hip-Hop, and Pop tend to attract older audiences, as indicated by their higher average ages.
- **Strategic implications:** This trend reveals how musical preferences shift with age, providing valuable data for platforms to:
 - Tailor recommendations based on user age.
 - Market specific genres more effectively to different age segments.
 - Curate age-appropriate playlists or experiences (e.g., younger-focused EDM festivals vs. older-audience R&B events).
- **Stakeholders:** Data analysts, marketing teams, and content curators.

```
6]: # Age versus genre
age_vs_genre = df.groupby(["Top Genre"])[ "Age"].mean().sort_values()
age_vs_genre
plt.figure(figsize=(8,5))
sns.barplot(x=age_vs_genre.index, y=age_vs_genre.values, palette="coolwarm")
plt.xticks(rotation=45)
plt.xlabel("Music Genre")
plt.ylabel("Average Age of listeners")
plt.title("Impact of Age on Music Preferences")
plt.show()
```



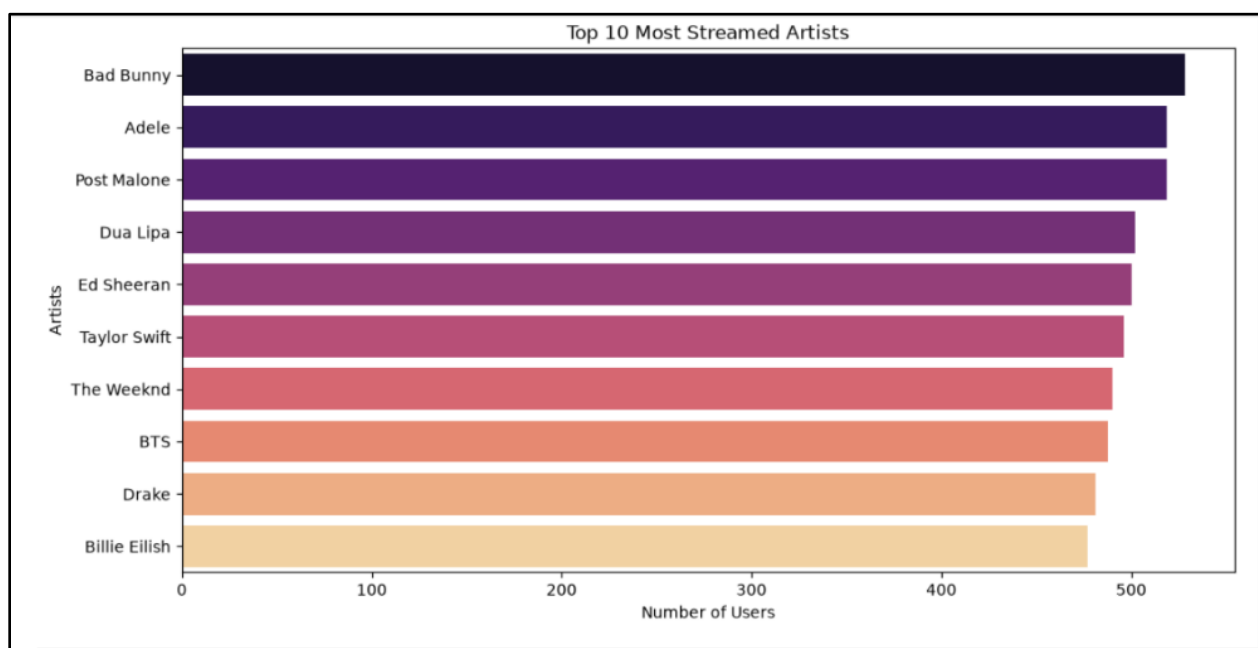
6. Top 10 Most Streamed Artists (Bar Chart)

- **Result:** Bad Bunny tops the chart as the most streamed artist among users. Adele and Post Malone closely follow in popularity. Other high-ranking artists include Dua Lipa, Ed

Sheeran, Taylor Swift, The Weeknd, BTS, and Drake. It shows listener loyalty to popular artists.

- **Strategic implications:** Enhances marketing strategies for artists and streaming services. It helps in designing artist-focused campaigns or feature spotlights in music streaming platforms.

```
[100]: # Top 10 Most Streamed Artists (Bar Chart)
plt.figure(figsize=(12,6))
sns.countplot(y=df["Most Played Artist"], order=df["Most Played Artist"].value_counts().index[:10], palette="magma")
plt.xlabel("Number of Users")
plt.ylabel("Artists")
plt.title("Top 10 Most Streamed Artists")
plt.show()
```



7. Regional Trends in Music Streaming Preferences (Bar Chart)

- **Result:**
 - Musical tastes vary significantly across regions, influenced by culture, media exposure, and local trends.
 - Reggae's broad reach across both western and non-western countries (USA, UK, India) suggests its global appeal and possibly rising popularity.
 - Rock remains strong in East Asian countries like South Korea and Brazil, pointing toward diverse and possibly niche rock fanbases.
 - Jazz and EDM maintain consistent popularity in European and North American regions, often associated with more urban and sophisticated music scenes.

- **Strategic implications:**

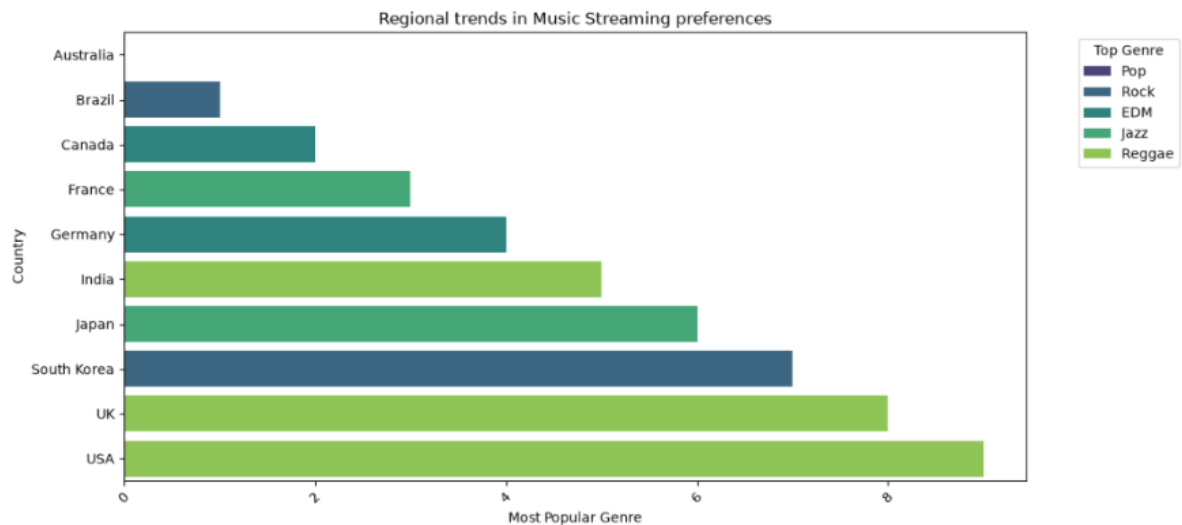
Streaming platforms can use this insight for:

- Localized recommendations,
- Region-specific playlists, and
- Genre-based promotions or partnerships.

Artists and labels can better target marketing and tour planning based on these regional genre preferences.

Most trending genre in the country

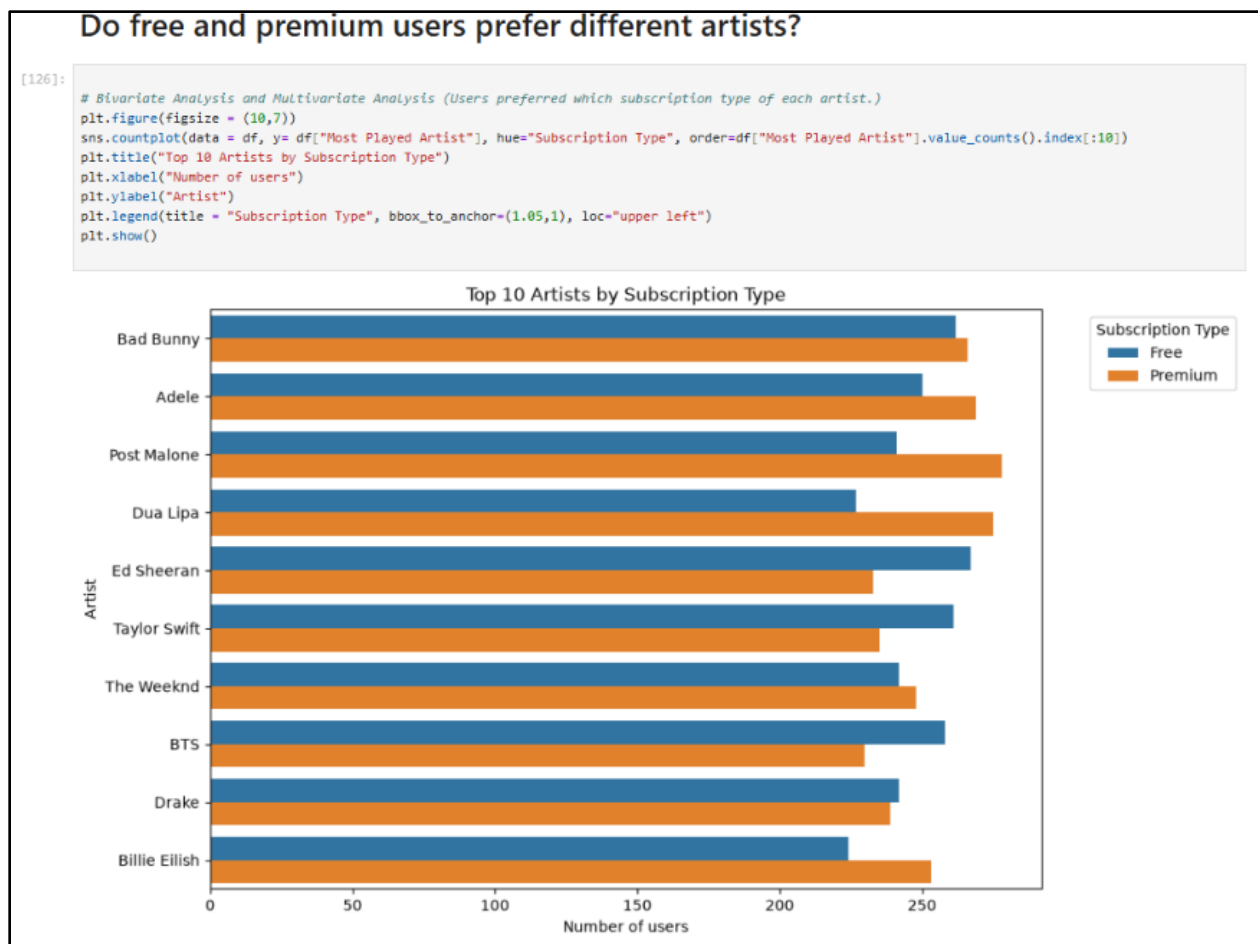
```
[124]: # Most trending genre in the country
regional_genre_trends = df.groupby("Country")["Top Genre"].agg(lambda x: x.value_counts().idxmax())
regional_genre_df = pd.DataFrame(regional_genre_trends).reset_index()
# sort data by country for better visualization
regional_genre_df = regional_genre_df.sort_values(by="Country")
# Plot a bar chart for regional trends
plt.figure(figsize = (12,6))
sns.barplot(y=regional_genre_df["Country"], x=regional_genre_df.index, hue=regional_genre_df["Top Genre"], dodge=False, palette="viridis")
plt.xlabel("Most Popular Genre")
plt.ylabel("Country")
plt.title("Regional trends in Music Streaming preferences")
plt.legend(title="Top Genre", bbox_to_anchor=(1.05,1), loc="upper left") # place legend outside
plt.xticks(rotation=45)
plt.show()
```



8. Top Artists by Subscription Type (Count Plot)

- **Result:**

- It indicates a higher engagement or preference for premium content listeners. It suggests that top artists are more popular among paying users, possibly due to access to exclusive content, better streaming quality, or ad-free experience.
- Artists like Dua Lipa, Post Malone, and Adele have a notably higher premium user base than free users.
- Ed Sheeran and BTS have more free users dominating the premium ones.
- **Strategic implications:**
 - Artists like **Dua Lipa, Adele, and Post Malone** could be **ideal for exclusive releases**, given their stronger paying listener base.
 - Streaming platforms might consider **targeted strategies for Free users** of BTS or Ed Sheeran to **convert them to Premium**.



9. Preferred Listening Time in each Country

- **Result:** Night is the most preferred listening time globally. UK users peak during night hours, suggesting a strong **evening engagement** culture (possibly post-work relaxation or leisure listening). Morning listening is popular in Japan and South Korea. These countries show a higher morning listenership compared to others, suggesting possible cultural or lifestyle influences like commuting or early routines.
- **Strategic implications:**
 - Platforms can push **targeted playlists or recommendations** at peak hours specific to each country (e.g., morning playlists in Japan, night mixes in the UK).
 - **Advertisers** can optimize placements for **maximum impressions** during regional peak listening hours.
 - Schedule global artist premieres to align with peak listening hours in each key country for maximum audience engagement.



10. Top Artist in Each Country by Repeat Song Rate

- **Results** : Adele is the most replayed artist in France and Japan, with France having the highest repeat rate overall (~50%). Ed Sheeran is most replayed in Canada, Germany, and South Korea, showing consistent popularity across diverse regions. **Taylor Swift** rules **Brazil**, while **Bad Bunny** leads in the **UK**, reflecting regional listening preferences.
- **Strategic Implications:**
 - Artists can target top-performing regions for concert tours or local partnerships.
 - Streaming platforms can personalize in-app recommendations based on repeat behavior patterns.

Question : Indentify which artist has the highest repeat song rate in each country

```
[127]: # Group by Country and Artist and calculate the average repeat song rate
country_artist_repeat_rate = df.groupby(["Country", "Most Played Artist"])[["Repeat Song Rate (%)"]].mean().reset_index()
# Find the top artist in each country based on the highest repeat song rate
top_artist_by_country = country_artist_repeat_rate.loc[country_artist_repeat_rate.groupby("Country")["Repeat Song Rate (%)"].idxmax()]
# sort the values for better readability
top_artist_by_country = top_artist_by_country.sort_values(by="Repeat Song Rate (%)", ascending = False)

# set figure size
plt.figure(figsize=(12,7))

# Create a barplot
sns.barplot(data=top_artist_by_country, x= "Repeat Song Rate (%)", y="Country", hue="Most Played Artist", palette = "coolwarm")

# Labels and title
plt.xlabel("Repeat Song Rate", fontsize=12)
plt.ylabel("Country", fontsize=12)
plt.title("Top Artist in each country by Repeat Song Rate", fontsize=12)
plt.legend(title= "Most Played Artist", bbox_to_anchor=(1.05,1), loc= "upper left")

# show the plot
plt.show()
```

